

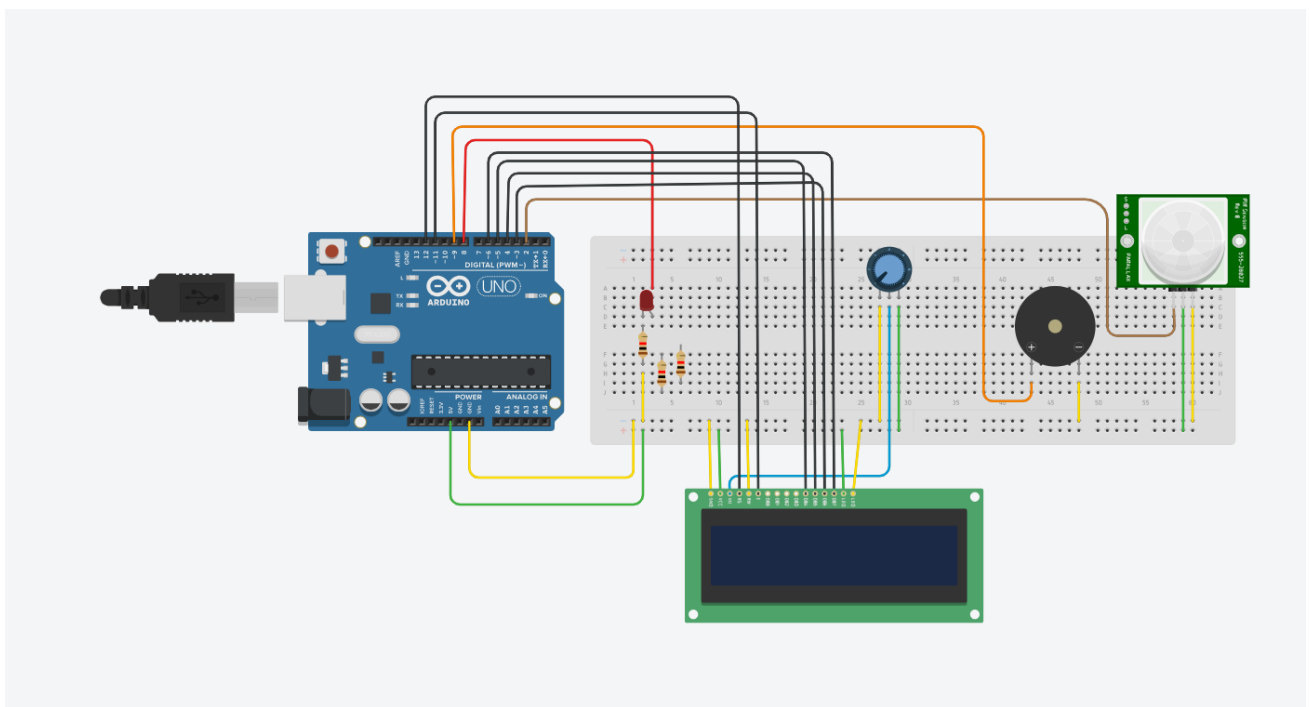
SecureSense : LCD-Based Smart Intrusion Detection and Alert System

SecureSense is an Arduino-based security system designed to detect unauthorized movement in a monitored area. The system uses a PIR (Passive Infrared) motion sensor to sense human movement by detecting changes in infrared radiation emitted by the human body.

When motion is detected, the Arduino processes the sensor signal and immediately activates:

- A buzzer to provide an audible alert.
- An LED to provide a visual warning.
- A 16×2 LCD display to show a motion detection message and the total number of detected motion events.

When no motion is present, the system remains in monitoring mode and displays a secure status message on the LCD. The intrusion counter helps track how many times movement has been detected, making the system more informative than a basic alarm.



Components Used

- Arduino Uno
- PIR Motion Sensor
- LCD 16×2 Display
- Potentiometer (LCD contrast adjustment)

- LED
- 220Ω Resistor
- Piezo Buzzer
- Breadboard
- Jumper Wires

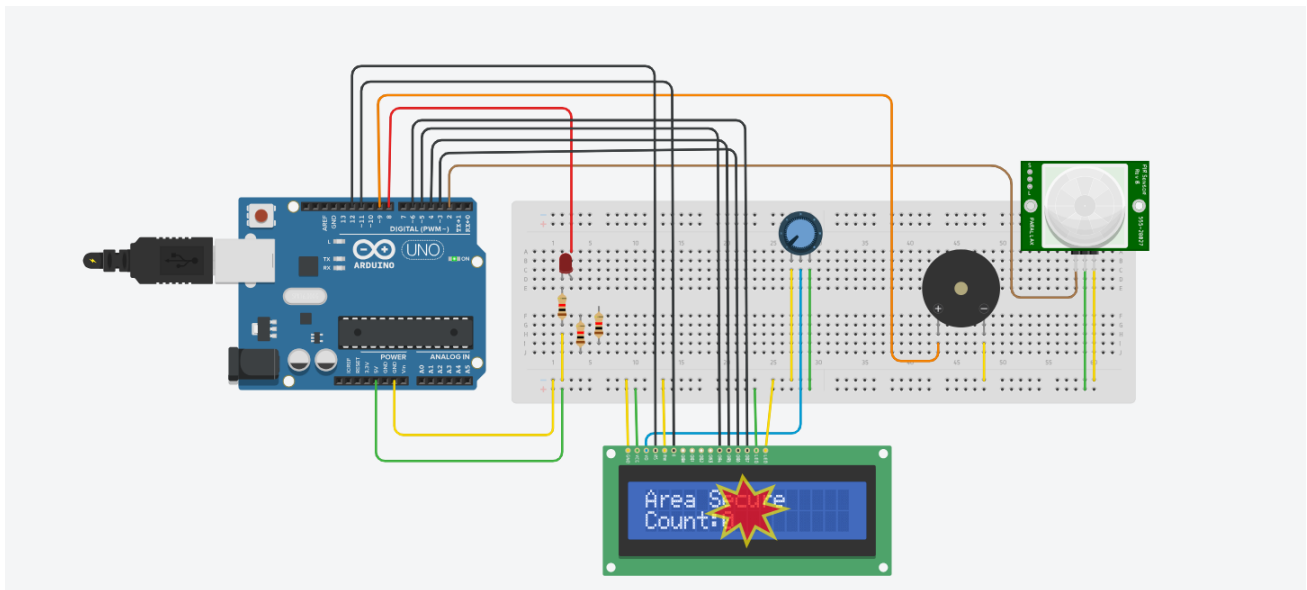
Working Principle

1. The PIR sensor continuously monitors the surroundings.
2. If no movement is detected:
 - LCD displays "**Area Secure**".
 - LED and buzzer remain OFF.
3. When movement is detected:
 - PIR sensor sends a HIGH signal to the Arduino.
 - Arduino turns ON the LED.
 - Arduino activates the buzzer.
 - LCD displays "**Motion Found!**".
 - Motion count is increased and displayed.
4. After the alert, the system returns to monitoring mode.

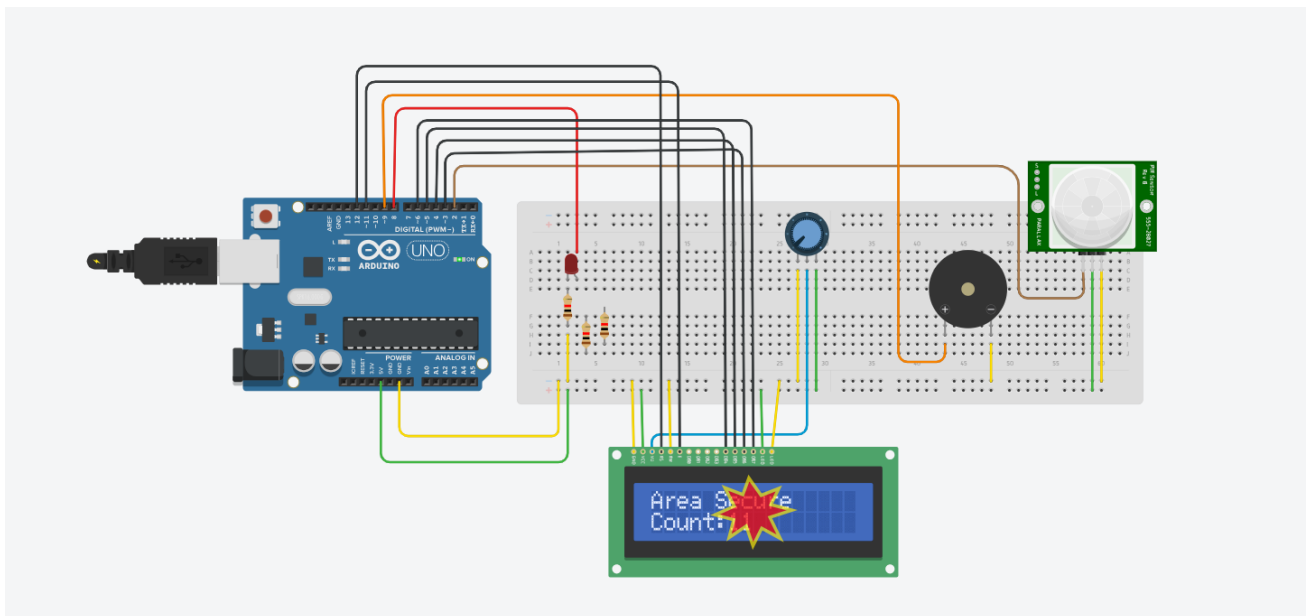
Applications

- Home security systems
- Office and workplace monitoring
- Restricted-area surveillance
- Museum and exhibit protection
- Smart building security

If no motion is detected



If motion is Detected, the count increases



Source Code:

```
// SecureSense: LCD-Based Smart Intrusion Detection and Alert System
```

```
#include <LiquidCrystal.h> // Include LCD library
```

```
LiquidCrystal lcd(12, 11, 5, 4, 3, 6); // LCD pin connections: RS, E, D4, D5, D6, D7
```

```
// Define component pins
```

```
const int pirPin = 2;    // PIR sensor output pin
```

```
const int ledPin = 8;    // LED pin
```

```

const int buzzerPin = 9; // Buzzer pin

int motionCount = 0; // Variable to store motion detection count

void setup() {

    pinMode(pirPin, INPUT); // Set PIR sensor as INPUT

    pinMode(ledPin, OUTPUT); // Set LED as OUTPUT

    pinMode(buzzerPin, OUTPUT); // Set Buzzer as OUTPUT

    lcd.begin(16, 2); // Initialize LCD (16 columns, 2 rows)

    // Display startup message

    lcd.setCursor(0, 0);

    lcd.print("SecureSense");

    lcd.setCursor(0, 1);

    lcd.print("System Armed");

    delay(2000); // Wait 2 seconds

    lcd.clear(); // Clear LCD

}

void loop() {

    int motion = digitalRead(pirPin); // Read PIR sensor value

    if (motion == HIGH) { // Check if motion is detected

        motionCount++; // Increase motion count

        digitalWrite(ledPin, HIGH); // Turn ON LED

        tone(buzzerPin, 1000); // Activate buzzer at 1000Hz

        lcd.clear(); // Clear LCD before displaying alert

        // First row message

        lcd.setCursor(0, 0);

        lcd.print("Motion Found!");

        // Second row message

        lcd.setCursor(0, 1);

        lcd.print("Count:");

        lcd.print(motionCount);

        delay(1000); // Keep alarm active for 1 second

        noTone(buzzerPin); // Stop buzzer

    }

```

```
else {  
    digitalWrite(ledPin, LOW);    // Turn OFF LED  
    noTone(buzzerPin);    // Ensure buzzer is OFF  
    lcd.clear();    // Clear LCD  
    // Display secure status  
    lcd.setCursor(0, 0);  
    lcd.print("Area Secure");  
  
    // Show total motion count  
    lcd.setCursor(0, 1);  
    lcd.print("Count:");  
    lcd.print(motionCount);  
    delay(500);    // Refresh every 0.5 seconds  
}  
}
```